Flaxseed Supplementation (not Dietary Fat Restriction) Reduces Prostate Cancer Proliferation Rates in Men Presurgery

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Background Prostate cancer affects one-out-of-six men during their lifetime. Dietary factors are postulated to influence the development and progression of prostate cancer. Low-fat diets and flaxseed supplementation may offer potentially protective strategies. Methods We undertook a multi-site, randomized controlled trial to test the effects of low-fat and/or flaxseed-supplemented diets on the biology of the prostate and other biomarkers. Prostate cancer patients (n=161) scheduled at least 21 days before prostatectomy were randomly assigned to one of the following arms: 1) control (usual diet); 2) flaxseed-supplemented diet (30 g/day); 2) low-fat diet (<20% total energy); or 4) flaxseed-supplemented, low-fat diet. Blood was drawn at baseline and prior to surgery and analyzed for prostate specific antigen (PSA), sex hormone binding globulin, testosterone, insulin-like growth factor-1 and binding protein-3, c-reactive protein, and total and low density lipoprotein cholesterol. Tumors were assessed for proliferation (Ki-67, the primary endpoint) and apoptosis. Results Men were on protocol an average of 30 days. Proliferation rates were significantly lower ($P < 0.002$) among men assigned to the flaxseed arms. Median Ki-67 positive cells/total nuclei ratios (x100) were 1.66 (flaxseed-supplemented diet) and 1.50 (flaxseed-supplemented, low-fat diet) vs. 3.23 (control) and 2.56 (low-fat diet). No differences were observed between arms with regard to side effects, apoptosis, and most serological endpoints; however, men on low-fat diets experienced significant decreases in serum cholesterol ($P=0.048$). Conclusions Findings suggest that flaxseed is safe, and associated with biologic alterations that may be protective for prostate cancer. Data also further support low-fat diets to manage serum cholesterol.